



ENGINEERING NOTEBOOK

Duck Rescue Engineering Challenge

Engineering Firm / Team Name	Duckez
Student Name(s)	Mia, Ignacio
Class / Period	6
Date Started — Date Completed	Aug 25 - Sept 4

How To Use This Notebook

- Fill in a section every day — each one is tagged with the Design Cycle stage it documents.
- Sketch and write in pen or pencil directly on these pages, or insert photos/CAD screenshots where noted.
- Be specific: “it didn't work” is not evidence. “The cup fell off after 2 attempts because the tape lost grip when wet” is.
- This notebook is graded using the Notebook & Reflection criterion of the Summative Rubric — see the Assessment packet.

The Design Cycle

Define Problem → Generate Concepts → Develop Solutions → Construct and Test → Evaluate Solution → Present Solution — and back to Define Problem if needed. Great engineers move through these stages more than once.

DEFINE PROBLEM

GENERATE CONCEPTS

DEVELOP SOLUTIONS

CONSTRUCT AND TEST

EVALUATE SOLUTION

PRESENT SOLUTION

SECTION 1

Problem & Goal

DESIGN CYCLE STAGE: DEFINE PROBLEM

What problem are you trying to solve? (Restate the Duck Rescue challenge in your own words.)

Write 2–3 sentences.

From an annual rubber duck race, many ducks are stuck in the water in a restricted area of the course. This area is where workers cannot go into the water to retrieve the rubber ducks, and so the ducks are stuck in the water. The issue is that we need to retrieve the ducks from the water safely from a distance of at least 3 feet away in under 2 minutes.

Design Requirements (constraints your device must meet)

- Distance: 3 ft
- Budget: N/A
- Materials: 1 net, 4 pvc tubes, 4 cross connectors, 3 corner connectors, 1 t connector, 4 cups, 10 feet of string, a roll of tape
- Time limit: 2 mins

What does “success” look like for your device?

Success looks like getting all of the rubber ducks back safely from a distance of at least three feet away in under 2 minutes. This is crucial because we cannot go into the restricted area of the water where the ducks currently are. Similarly, if the ducks aren't rescued in under 2 minutes, they will drown?

SECTION 2

Research & Inspiration

DESIGN CYCLE STAGE: GENERATE
CONCEPTS

Insert photos, screenshots, or notes from research (existing grabbers, cranes, nets, rescue tools):

Photo / sketch space



What did you learn from your research? (1–2 bullet points)

- We can use a pole to reach farther than our arms can
- It doesn't need to be extravagant to work

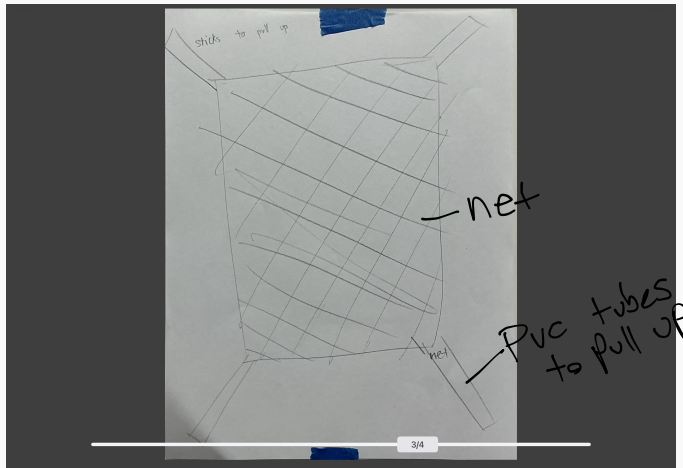
SECTION 3

Initial Ideas

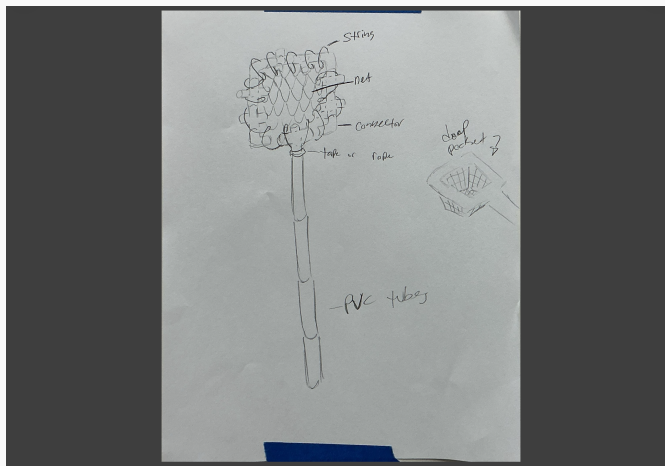
DESIGN CYCLE STAGE: DEVELOP SOLUTIONS

Sketch and label at least 3 different design ideas. For each, note how it works and one risk.

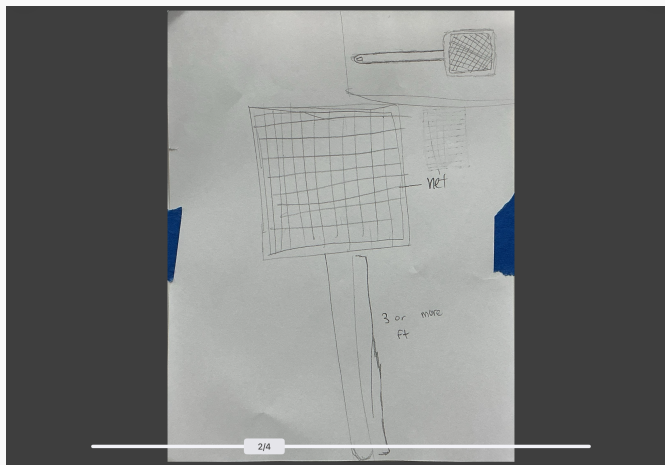
Idea 1



Idea 2



Idea 3



Which idea did your team choose, and why?

Our team chose idea 1, because it covered the most space. Therefore it would collect the most ducks in one and was the most original compared to everyone else's designs.

SECTION 5

Prototype #1

DESIGN CYCLE STAGE: CONSTRUCT AND
TEST

Insert photos of your prototype:

Photo space



What worked?

It was the width of the pool, and we were able to get it over the pool from both sides from outside the boundary lines. The extension strings connected to the handles worked to extend the arms reach, and nothing got detached or broke during the testing.

What needs improvement?

The net was unable to break the water's surface, resulting in collecting zero ducks. The extension strings also got tangled before testing, resulting in having to fix it before.

SECTION 6

Testing — Round 1

DESIGN CYCLE STAGE: CONSTRUCT AND
TEST

Record your Round 1 test data (full log is in your Student Portfolio):

Attempt #	Ducks Reached	Ducks Collected	Issues Observed
1	1	0	Net cannot break Water Surface
2			
3			

What failed, and why? (evidence, not guesses)

Our design failed, because it was unable to break the water's surface. This was a flaw in our prototype, because to rescue all of the ducks, the net must go under the water and scoop or lift to above to carry the ducks out.

SECTION 7

Iteration & Changes

DESIGN CYCLE STAGE: EVALUATE
SOLUTION

Insert updated photos or CAD screenshots of Prototype #2:

Photo space



Change Made	Reason (evidence from testing)
Added weights to the net	So net can break water surface and go under ducks
Secured by tying	Instead of tape, so it doesn't fail by getting wet

SECTION 9

Reflection & Presentation

DESIGN CYCLE STAGE: PRESENT
SOLUTION

Summarize your final design and results in 2–3 sentences — as if explaining it to a client.

The 'Duckez' duck safety rescue net is a contraption to get toys and ducks out of pools from a distance of 3 feet away. Grips on the end of poles make for easy handholds to use the product; Weights on the end of the net breaking the water surface.

What was successful?

Our revision to add weights was successful. From our first attempt and rescuing 0 ducks, to our last attempt of collecting more, the addition proved to be successful. Similarly, the extension strings were also successful. They were not in our original design.

What would you improve with more time?

With more time, I would move the upper weight to on the edge of the net. This would make it a more even scoop, resulting in collecting more ducks. I would also re-tie the extension strings on so they would not break.

What engineering skills did you use? (circle all that apply)

CAD · Prototyping · Testing · Data Analysis · Budgeting · Communication · Teamwork

Which Design Cycle stage was hardest for your team, and why?

For our team, teamwork was something I think we found the hardest. As the issue of the group, I am a major workaholic and likely bossy, too. These are all qualities I'm trying to work on for the future, so it doesn't become another issue in later projects.

